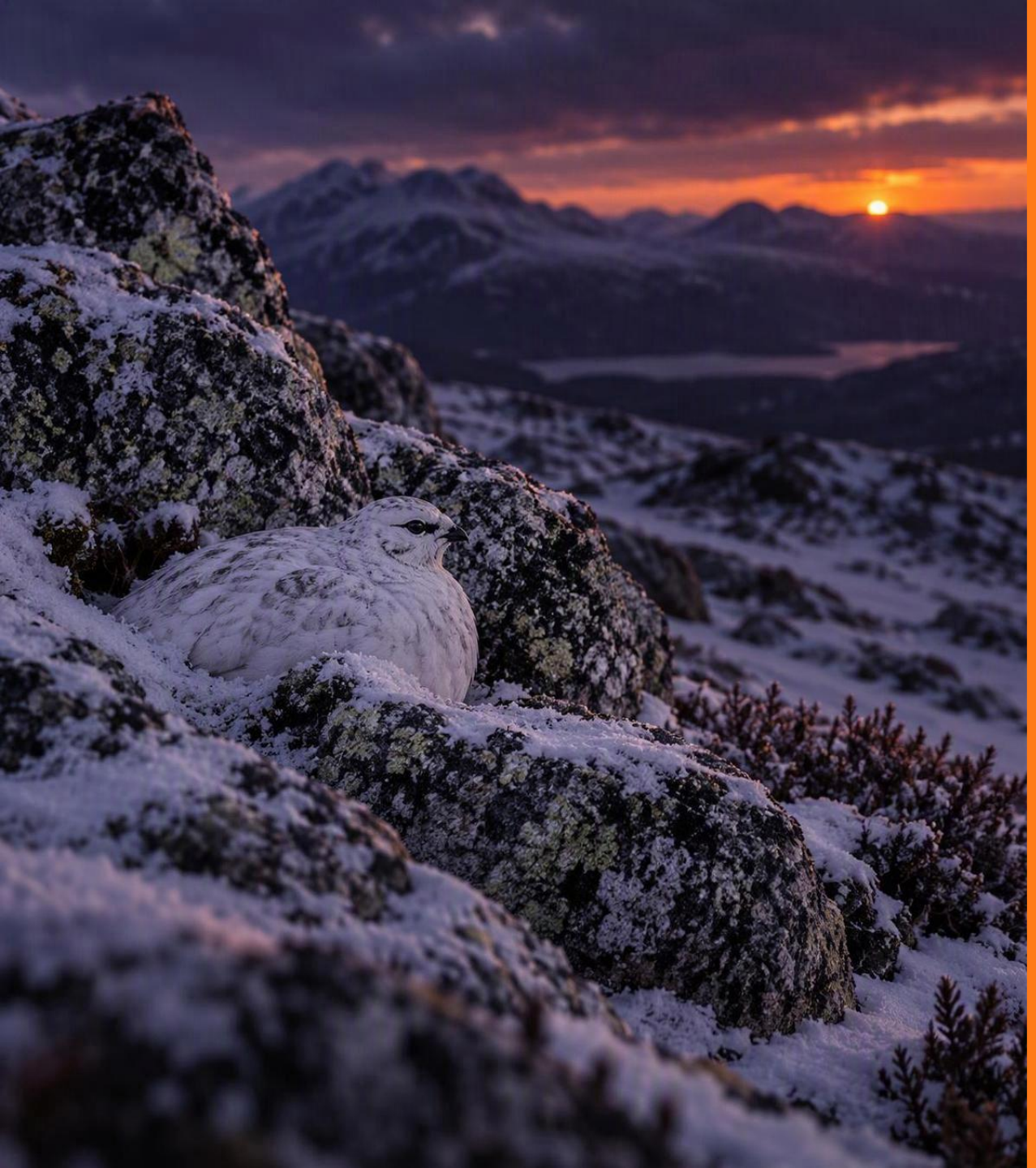
A TALE OF LEGACY IT

Do you have skeletons in the IT closet?

Legacy servers, Azure Arc, hybrid, Extended Security Updates, and the path to modernising

Sarah Lean | Sr Solution Engineer





A TALE FROM THE HIGHLANDS

Meet the ptarmigan: Scotland’s master of hiding in plain sight

Today’s hunt: what’s camouflaged in your IT estate?

WHO AM I?

Sarah Lean, aka Techielass

Sr Solution Engineer @ Microsoft

20 YEARS IN TECH

IT Pro at heart, from the server room to the cloud

COMMUNITY

Founder of the Glasgow Azure User Group and STEM Ambassador

TECHIELASS

Blogging practical how-tos at www.techielass.com

A techie talking tech, not a sales pitch.



WHAT'S LURKING IN THE DARK

Four skeletons we find in almost every closet

01

Unpatched servers

Out-of-support OS versions no longer receive regular security updates.

02

Forgotten databases

Old SQL Server instances still powering business-critical apps.

03

Invisible inventory

Servers spread across datacentres, edge and other clouds with no single view.

04

Compliance ghosts

Unsupported software raises audit, regulatory and insurance risk.



THE CLOCK IS TICKING

Support deadlines are creeping up on the estate

14 Jul 2026

SQL Server 2016

End of support

13 Oct 2026

Windows Server 2012 / R2

Final ESU update

12 Jan 2027

Windows Server 2016

End of support

12 Jul 2027

SQL Server 2014

Final year of ESUs ends

Source: Microsoft Lifecycle & ESU guidance (Microsoft Learn, 2026)



WEIGHING UP THE OPTIONS

What are your options for legacy servers?

01

Decommission

Retire servers and apps that no longer earn their keep. No risk, no licence, no running cost.

QUICK WIN

02

Run at risk

No more security patches. Every new vulnerability stays open, and compliance and audit exposure grows.

HIGHEST RISK

03

Buy ESUs

Pay for up to three years of critical security updates. A paid bridge that buys time, not a cure.

STOP-GAP

04

Modernise

Upgrade in place, migrate to Azure, or replatform. Removes the deadline and ends ESU spend.

LONG-TERM FIX



THE BRIDGE, NOT THE DESTINATION

Extended Security Updates buy time, not a cure

3

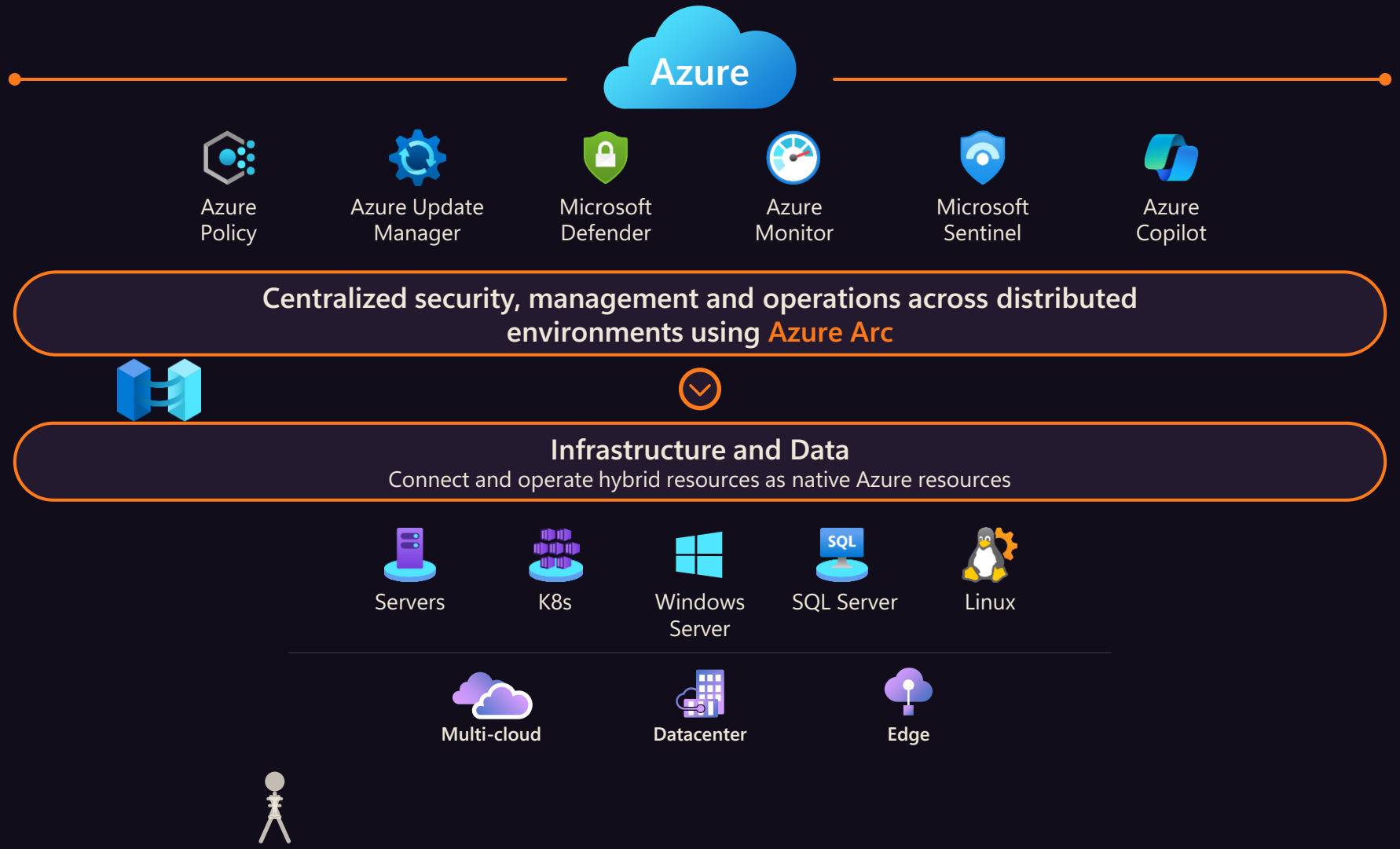
years maximum of ESU cover after end of extended support

- **What you get**
  Critical (and, for Windows Server, Important) security updates only.
- **What you don't**
  No new features, no non-security fixes, no general technical support.
- **Why it matters**
  ESUs are a last-resort, paid bridge while you upgrade or migrate.
- **When to stop**
  ESU charges only stop once you modernise.



AZURE ARC

One control plane for every hidden server



AZURE ARC

Shine a torch into every corner of the estate

Azure Arc connects servers in your datacentre, at the edge and in other clouds to Azure, so you can manage them as Azure resources.

See

Inventory, tag and query every server in one place with Azure Resource Graph

Govern

Apply Azure Policy and role-based access consistently, everywhere

Secure

Extend Microsoft Defender for Cloud to on-premises machines

Patch

Orchestrate updates, including ESUs, with Azure Update Manager

Inventory and core management in the Arc control plane are free; services like Defender and ESUs are billed separately.



ESUS ENABLED BY AZURE ARC

A friendlier way to keep the ghouls out

Classic ESUs

- Bought via Volume Licensing
- Purchased in yearly increments
- Keys managed server by server
- Paid for the full year regardless

ESUs via Azure Arc

- Pay-as-you-go monthly subscription
- Enrol and license from the Azure portal
- Patch centrally with Azure Update Manager
- Stop paying as servers upgrade or modernise

Remember: on-premises or in Azure, ESUs are a paid bridge. The real saving comes from modernising.



HOW ESUS ARE PRICED

Think of ESUs as a meter that runs per core

Cores
Licensed per core

×

Edition
The edition you run

×

Months on ESU
Every month adds up

=

Your ESU bill
Stops when you modernise

Windows Server 2016

ESUs from 12 Jan 2027

- Priced per physical core, like the OS licence
- Standard or Datacenter edition sets the rate
- Covers the operating system only
- Minimum 8 cores per VM, 16 per physical server

SQL Server 2016

ESUs from 14 Jul 2026

- Per core: each VM's vCores, or the host's physical cores
- Standard or Enterprise edition sets the rate
- Charged on top of any Windows Server ESUs
- Minimum 4 cores per VM or physical processor

Two meters, not one: a SQL 2016 box on Windows Server 2016 needs both ESUs. Enrol late and you're back-billed to end of support.



OPTION 04: MODERNISE

Three ways to switch off the ESU meter

01

Upgrade in place

Upgrade the OS or SQL Server on the same box. Quickest route, but you keep the old hardware and years of drift.

QUICKEST

02

Side-by-side

Build a new server on a supported version, move the role or data, then cut over. The old box is your rollback.

CLEANEST

03

Replatform

Move the workload to a managed service, such as Azure SQL Managed Instance. The OS stops being your problem.

LEAST TO RUN



WHY REPLATFORM?

Why replatforming is worth the effort

01

Less to attack

No OS to harden, no RDP or SMB to expose, no local admin accounts. Entra ID auth, encryption and auditing built in.

SECURITY

02

Off the treadmill

An upgrade just resets the support clock. On a managed service, OS and engine patching is the platform's job.

LIFECYCLE

03

Less toil

Backups, HA and patching are built in. The server only one person understands stops being a single point of failure.

OPERATIONS

04

Data you can use

AI projects need data that's current, reachable and governed. Data on a forgotten 2016 box is none of those.

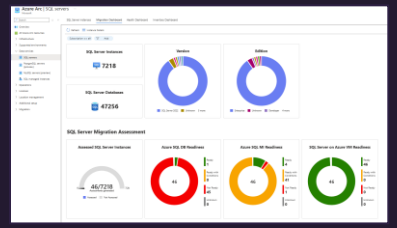
AI READINESS



REPLATFORM IN PRACTICE

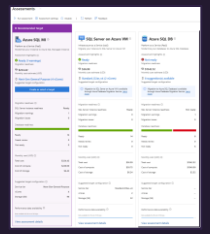
Moving SQL Server off the ESU meter

At scale Discovery



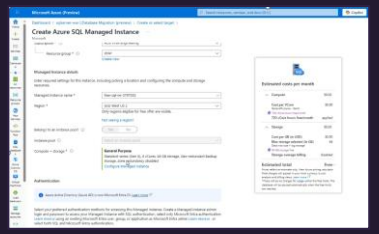
- At Scale Discovery
- Estate level inventory view

Continuous Assessment



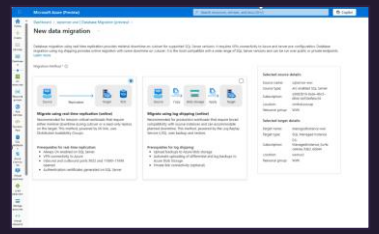
- Built-in and Automatic
- Migration Readiness
- Right-sized SKU recommendation
- Cost projections

Simplified Provisioning



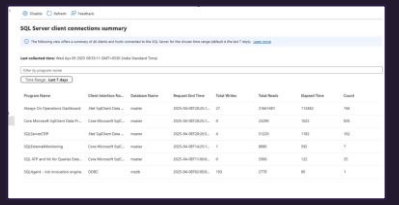
- Provision any Azure SQL target inline
- Free offerings included

Real-time Replication



- Realtime replication
- Log shipping and Native restore
- Supports MI and VM targets

Confident Cutover



- Inbound client connections
- Ensure transition to the new target
- Cutover with a click!

The data moves in days. Testing, sign-off and connection strings take the time.

WHEN THE SKELETON IS THE APP

Sometimes the server isn't the problem

01

Test it on a newer OS

Many .NET Framework 4.x apps run unchanged on Windows Server 2022 or 2025. Prove it with a side-by-side build before anyone plans a rewrite.

TRY THIS FIRST

02

Map the dependencies

Find what talks to what before you move anything. One app is often a web tier, a database, a file share and a scheduled task.

BEFORE YOU MOVE

03

Bring in the developers

If the code must change, hand over a head start: a code assessment, the upgrade blockers, and AI-assisted tooling such as GitHub Copilot.

WHEN CODE MUST CHANGE

Your role: you can't fix the code, but you can start the conversation with the people who can.



CLEARING OUT THE CLOSET

Four steps to get out of the haunted house

STEP 1

Discover

Arc-enable everything, build a real inventory, and switch off what nobody owns

STEP 2

Assess

Map dependencies, find the app blockers, and pick a path for each workload

STEP 3

Protect

Put ESUs on what can't move yet, and patch it all from one place

STEP 4

Modernise

Upgrade, rebuild or replatform, then turn the ESU licence down

Start with what's exposed: 2012 first, then SQL 2016, then Windows Server 2016 before 12 Jan 2027.



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NO MORE SKELETONS

Let's turn the lights on together

1. Arc-enable every server
2. Run an inventory query
3. Build an ESU and modernisation roadmap

GRAB THE SLIDES & RESOURCES

aka.ms/skeletons

Questions?

